EXERCISE: LOG FILE ANALYSIS FOR INCIDENT DETECTION LAB

Aim

To analyze system log files and detect suspicious activities such as failed login attempts and identify suspicious IP addresses.

Objective

- Understand system log files
- Detect suspicious activities and incidents
- Analyze logs using Linux tools
- Identify attack patterns (failed logins, brute force, etc.)

Requirements

- Kali Linux / Ubuntu VM
- Basic Linux command knowledge
- Tools:
- Kali Linux
- `journalctl`
- `ssh`
- `grep`, `awk`, `sort`
- `grep`
- `awk`
- `less`
- `cat`
- `journalctl`

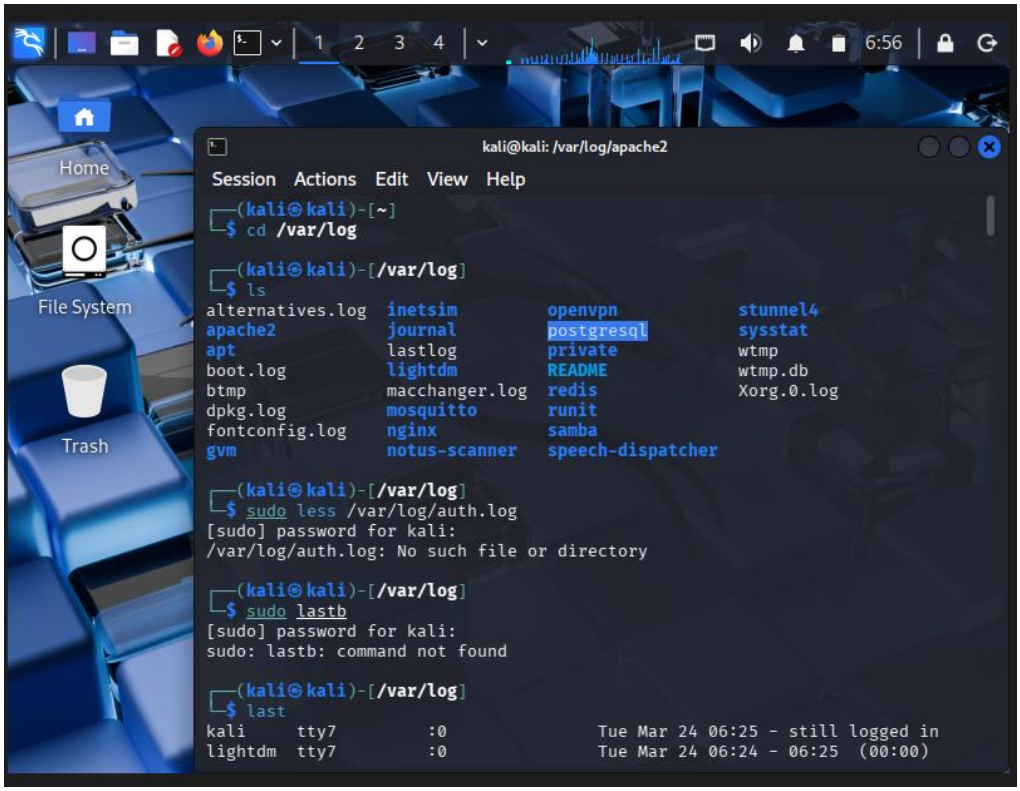
Understanding Log Files

Navigate to Log Directory

cd /var/log

ls

The `/var/log` directory contains system and application logs such as Apache, Nginx, and login records used for security analysis.



What Your Output Shows

From your `ls`:

✓ `dpkg.log` → `package activity`
✓ `apache2` → `web server logs`
✓ `nginx` → `web logs`
✓ `journal` → `system logs (IMPORTANT)`
✓ `wtmp, btmp` → `login records`

Successful Logins (wtmp)

Command:

last

Capture:

- Login history

This displays successful login sessions. It helps identify legitimate and suspicious user access.

System Logs using journalctl

Command:

journalctl | less

Capture:

- Logs output

Failed SSH Attempts

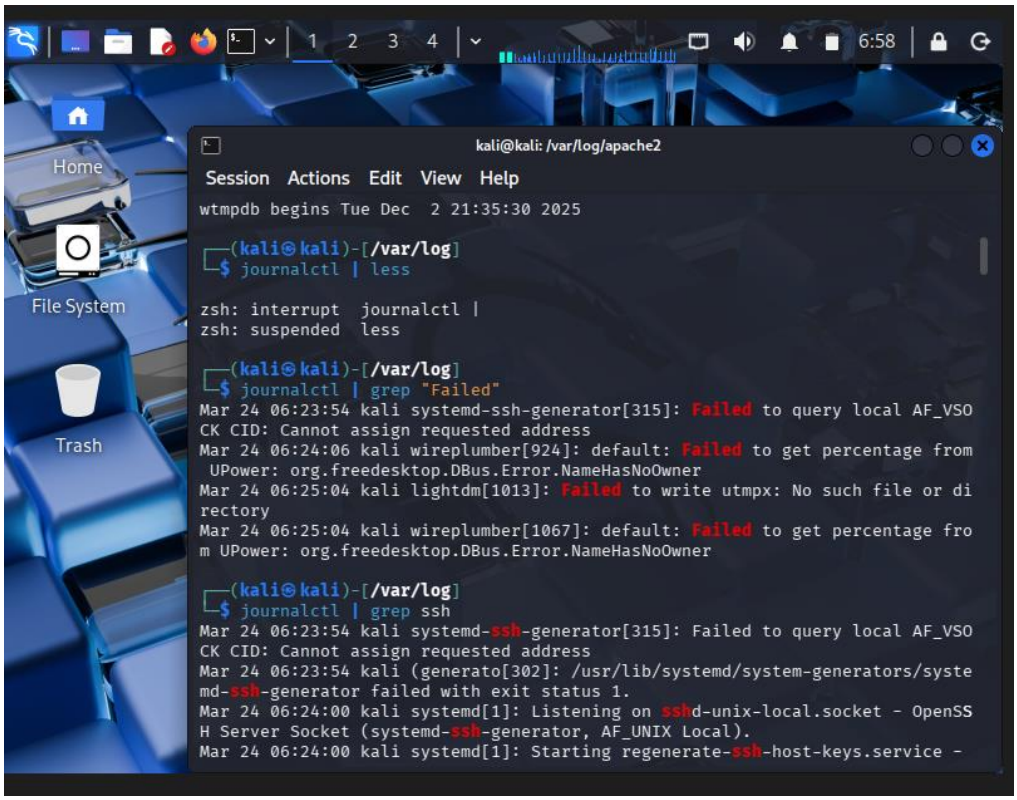
Command:

journalctl | grep "Failed"

SSH Login Activity

Command:

journalctl | grep ssh

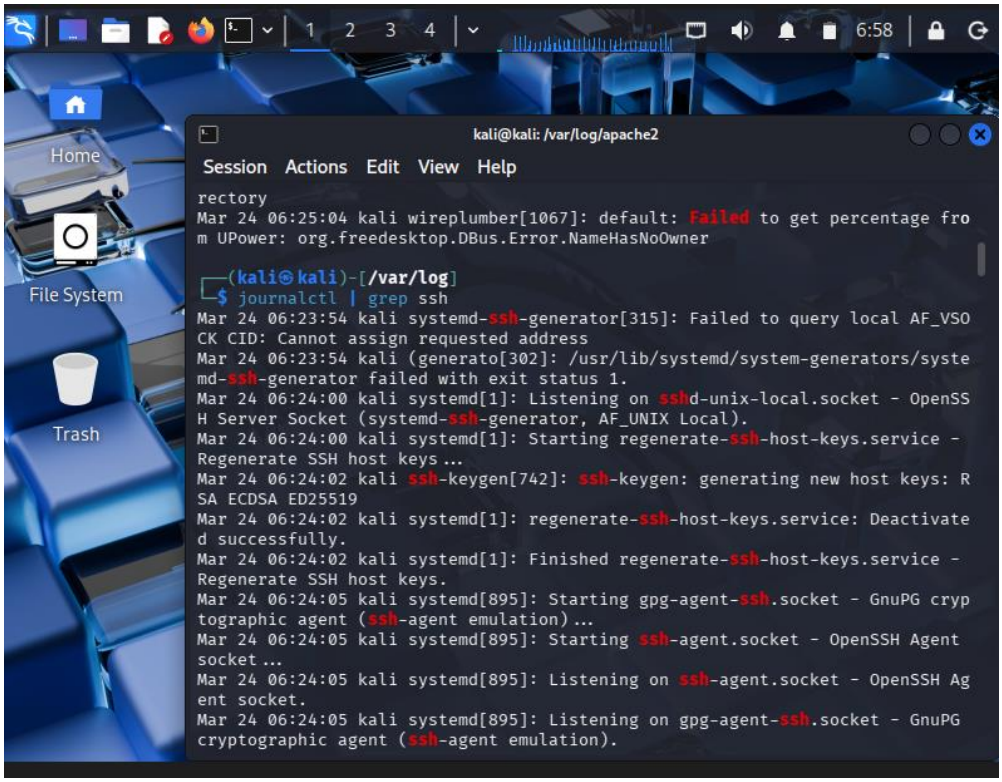


Explanation:

The journalctl command provides centralized system logs including authentication, services, and system events.

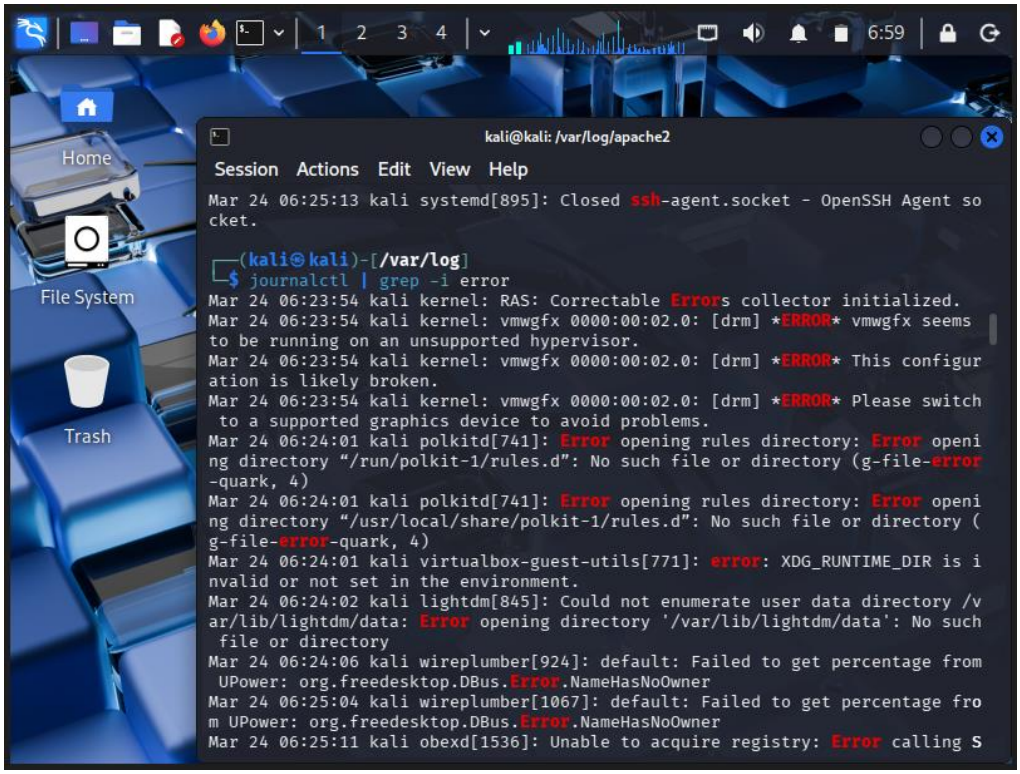
This screenshot shows failed authentication attempts, which may indicate unauthorized access attempts.

This screenshot also displays SSH-related logs, including login attempts and connections.



Error Detection

Command: journalctl | grep -i error



The above screenshot highlights system errors, which may indicate abnormal behavior or system issues.

Apache Log Analysis

Command:

```
cd /var/log/apache2
ls
```

Then:

```
sudo less access.log
```

Apache logs record web requests. These logs help identify suspicious web activity such as repeated requests or attacks.

Suspicious Requests

Command:

```
grep "404" /var/log/apache2/access.log
```

This shows failed web requests (404 errors), which may indicate scanning or probing attacks.

Package Activity

Command:

```
less /var/log/dpkg.log
```

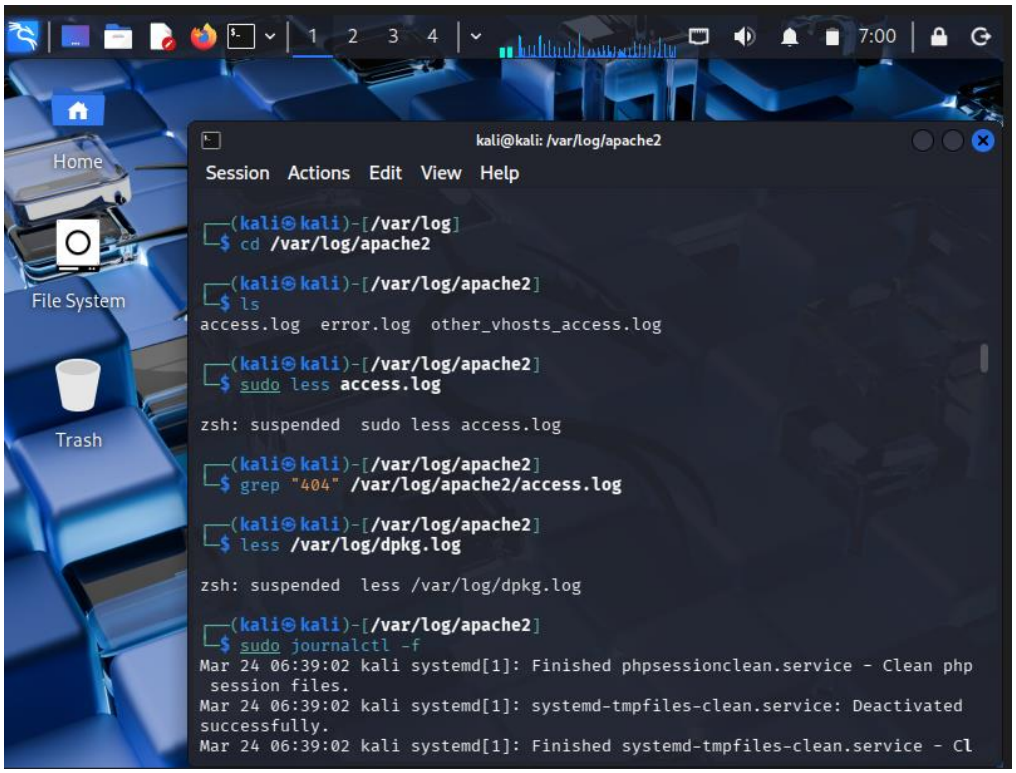
This shows software installation and update logs, useful for detecting unauthorized changes.

Real-Time Monitoring

Command:

```
sudo journalctl -f
```

This shows real-time system log monitoring, allowing detection of incidents as they occur.



QUICK CHECKLIST

Check	Command	Suspicious If
Failed logins	`lastb`	Many attempts
SSH activity	`journalctl	grep ssh`	
Login history	`last`	Odd timing
Errors	`journalctl	grep error`	
Web logs	`grep 404`	Many failed requests

STEP-BY-STEP ANALYSIS OF YOUR LOG

1. System Services (NORMAL)

```
Finished phpsessionclean.service
Finished systemd-tmpfiles-clean.service
```

✓ These are routine background tasks
✓ No issue

Regular system maintenance services were observed, indicating normal system operation.

2. CRON Jobs (NORMAL)

```
CRON: session opened for user root
(root) CMD (debian-sa1)
```

✓ Automated scheduled task
✓ Runs periodically

Scheduled cron jobs were observed, which are normal automated system processes.

3. Your Own SUDO Activity (NORMAL)

```
kali : COMMAND=/usr/bin/less access.log
kali : COMMAND=/usr/bin/journalctl -f
```

✓ This is YOU running commands
✓ Not suspicious

Sudo activity corresponds to legitimate administrative actions performed by the user.

4. XFCE / GUI Services (NORMAL)

```
xfconfd.service started
```

✓ Desktop environment service
✓ Normal

5. IMPORTANT PART

● Repeated Kernel Messages

```
dndHGCM: Received message HOST_DND_FN_GH_REQ_PENDING
dnd: No guest source window
```

These lines are:

```
Repeated MANY times
Coming from VirtualBox (VBoxDRM / DnD)
```

🔍 Is This Suspicious?

Technically:

- NOT an attack
- But **abnormal repetitive behavior**

❗ Why it matters:

- Repeated logs = unusual system activity
- Could indicate:
  - Misconfiguration
  - Resource issue
  - VirtualBox interaction problem

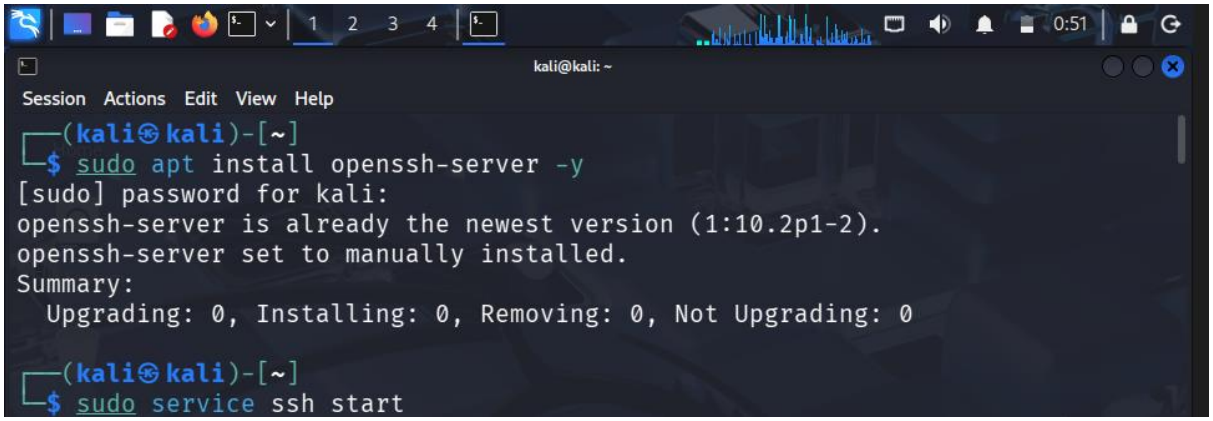
METHOD 1: SIMULATE FAILED SSH LOGIN

Step 1: Ensure SSH is Installed

```
sudo apt install openssh-server -y
```

Step 2: Start SSH Service

```
sudo service ssh start
```

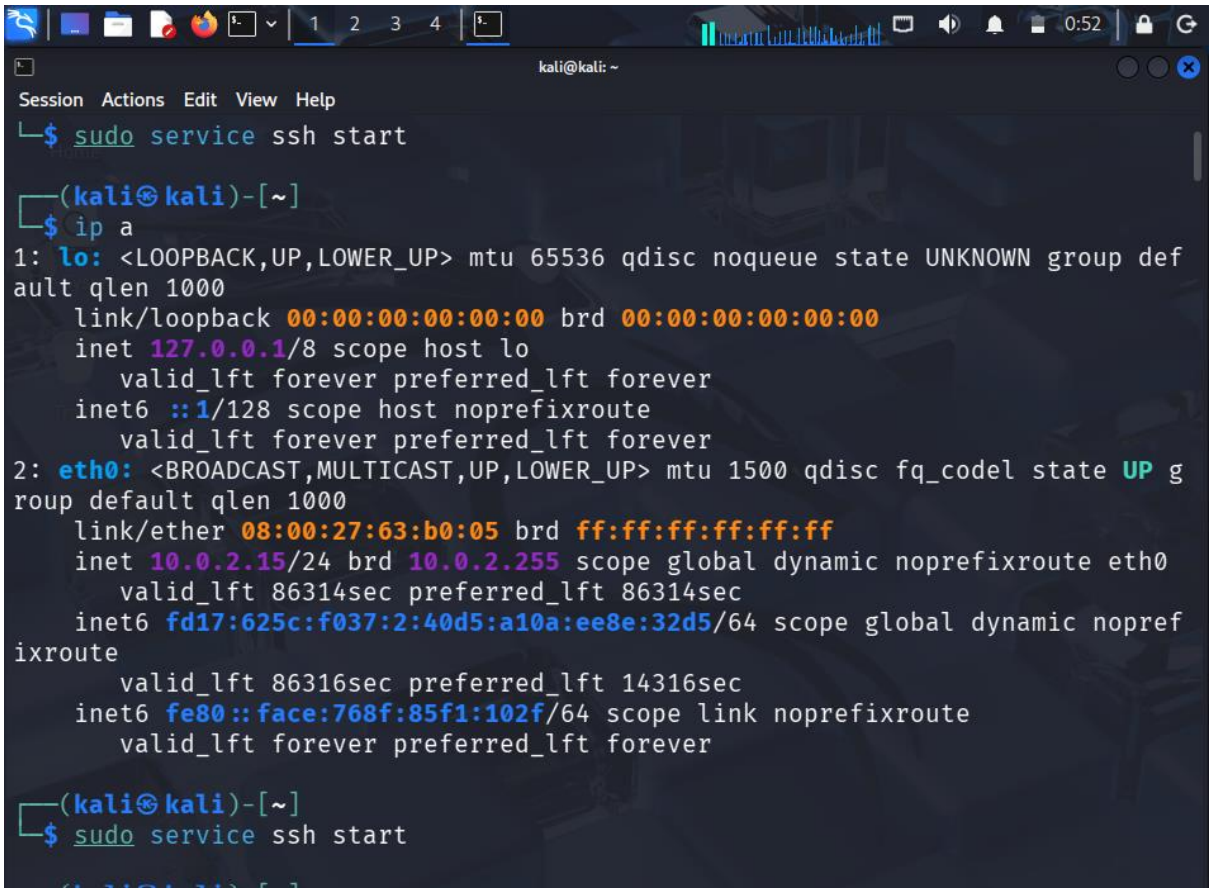


Step 3: Find Your IP Address

```
ip a
```

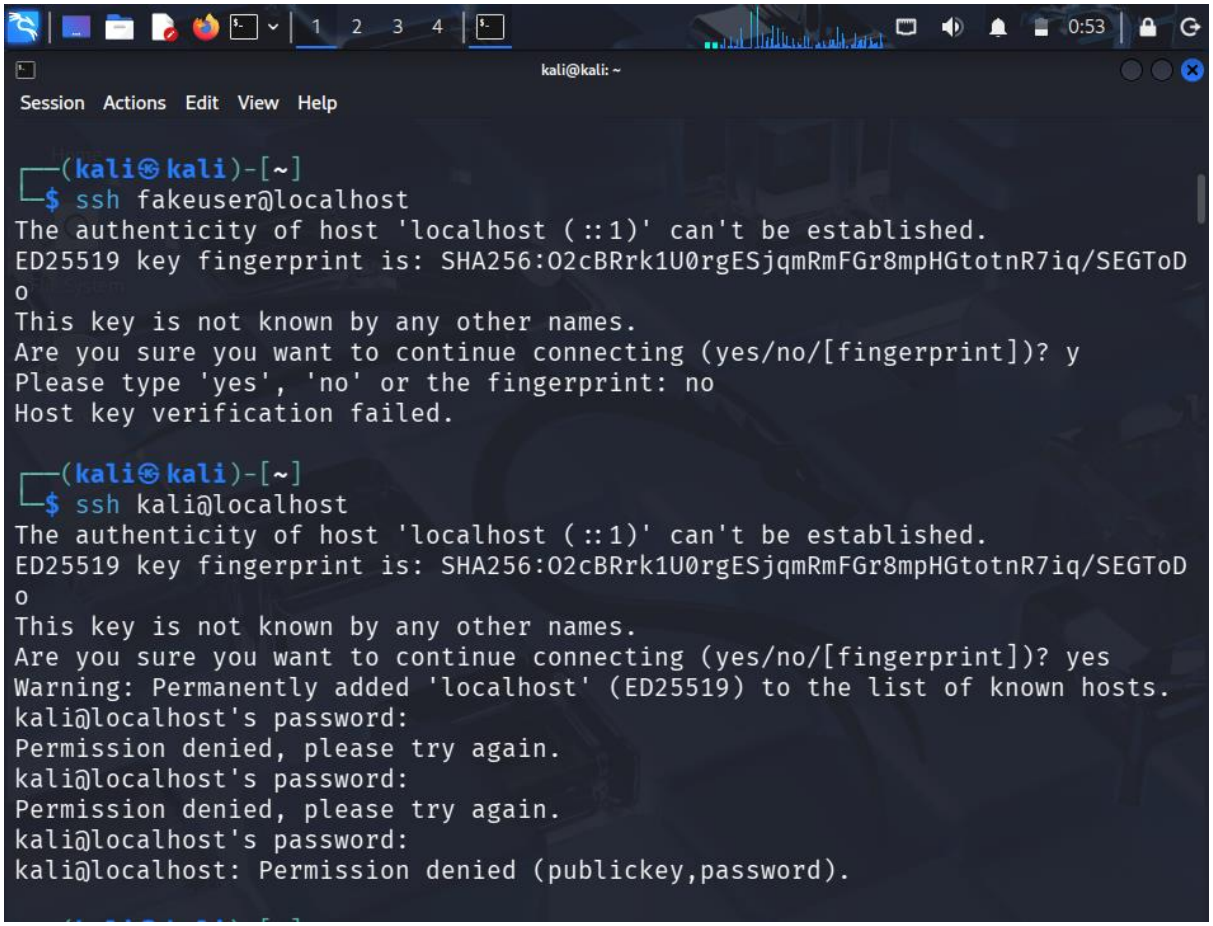
Look for something like:

inet 192.168.x.x



Step 4: Attempt Wrong Login (GENERATE FAILURES)

run:
ssh fakeuser@localhost
OR
ssh kali@localhost
Enter **wrong password multiple times (5–10 times)**



Now Analyze Logs

Step 5: Check Failed Logins

```
journalctl | grep "Failed password"
```

☞ You should now see:

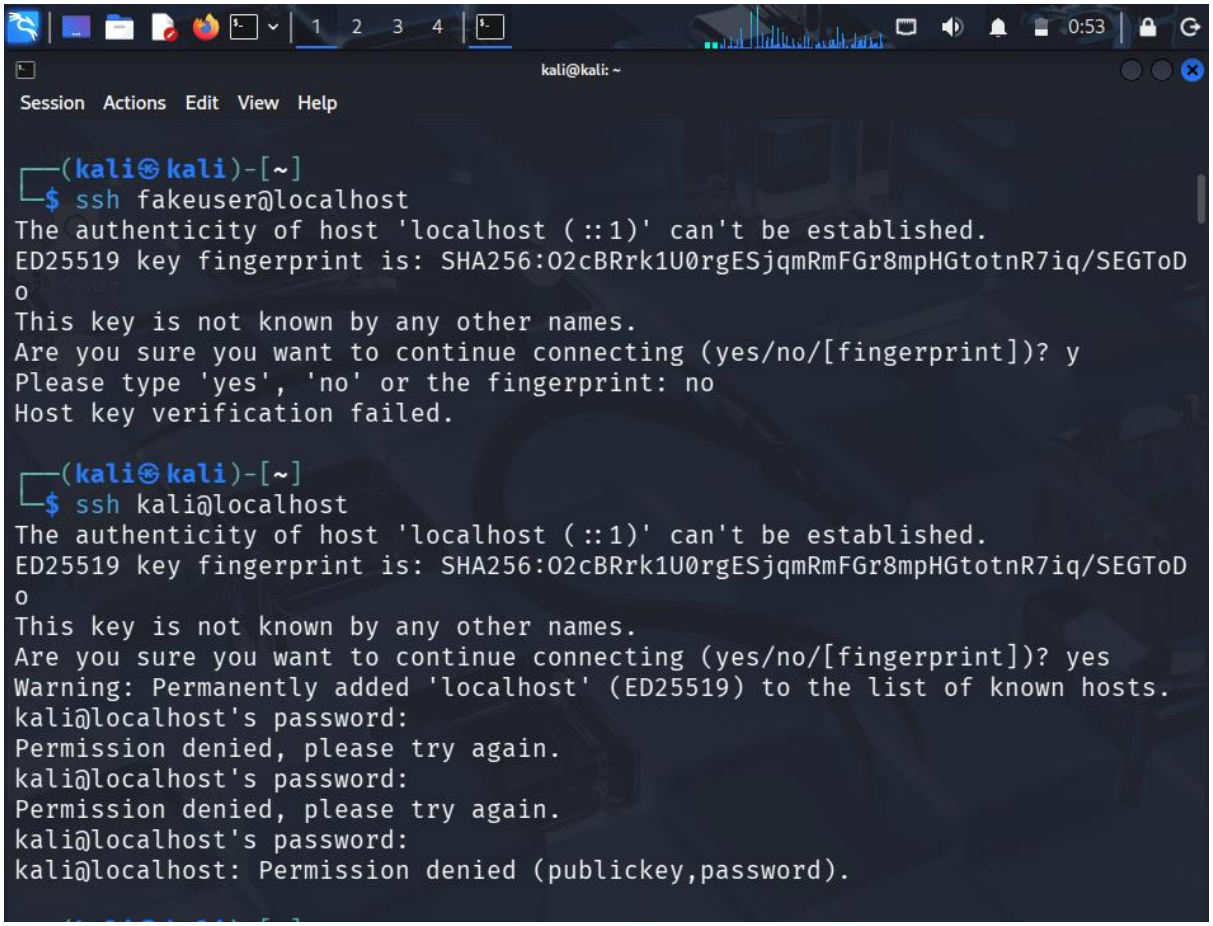
```
Failed password for kali from 127.0.0.1 port ...
```

Step 6: Extract Suspicious IP

```
journalctl | grep "Failed password" | awk '{print $11}'
```

Output:

```
127.0.0.1
```

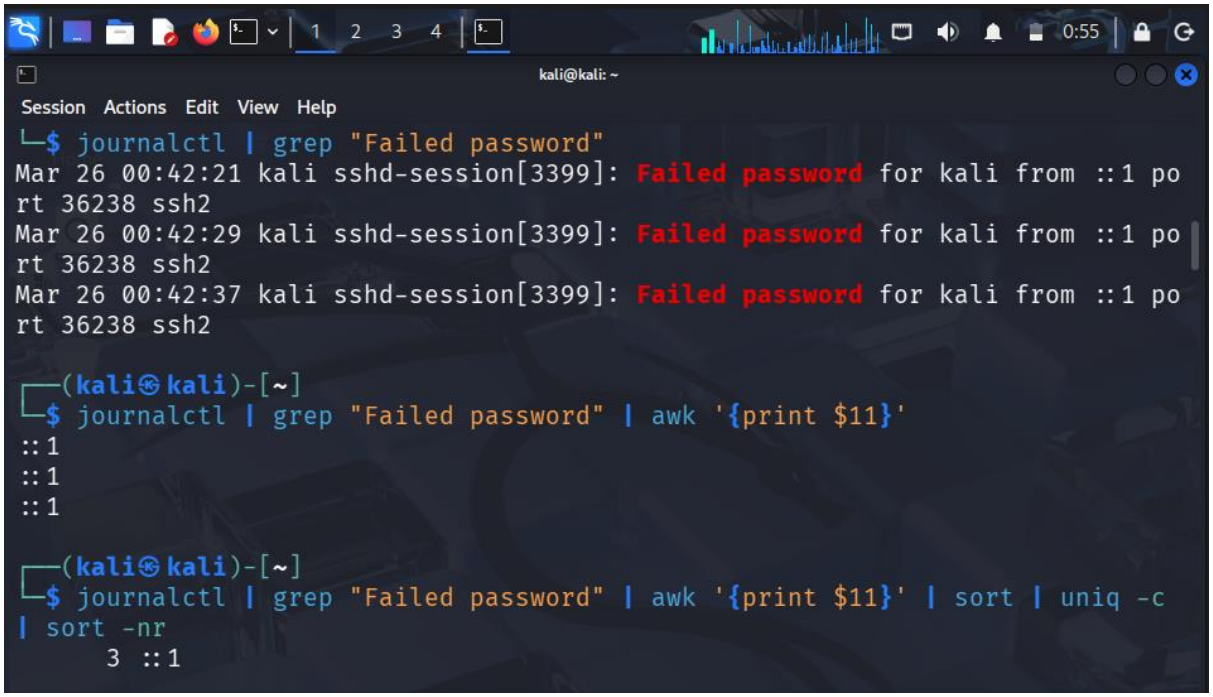


Step 7: Count Attempts Per IP

```
journalctl | grep "Failed password" | awk '{print $11}' | sort
| uniq -c | sort -nr
```

Example:

```
10 127.0.0.1
```



INTERPRETATION (VERY IMPORTANT)

Multiple failed login attempts were detected from IP address 127.0.0.1, indicating a simulated brute force attack.

METHOD 2: Use lastb (Alternative)

```
sudo lastb
```

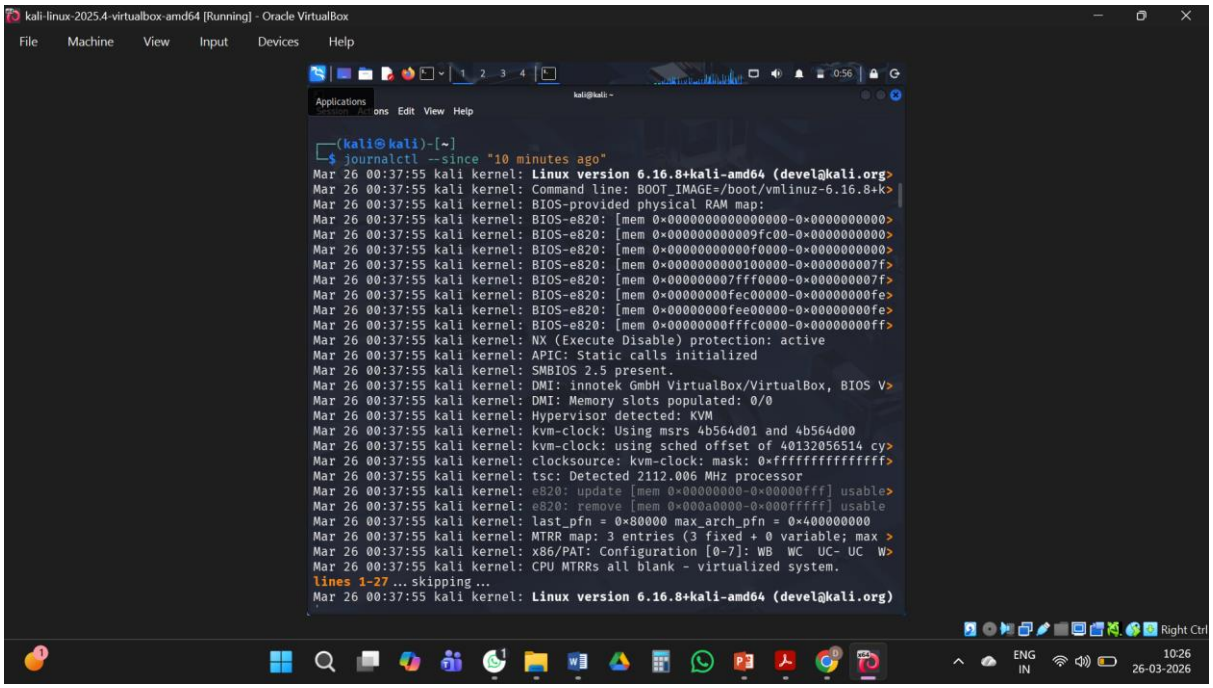
Shows failed login records

HOW TO IDENTIFY "SUSPICIOUS IP"

Simple rule:

Condition	Meaning
Same IP repeated many times	🚨 Suspicious
Unknown IP	🚨 Suspicious
High frequency attempts	🚨 Brute force

Run:

`journalctl --since "10 minutes ago"`

Shows recent attack activity